

Markdown 템플릿

수강생이 GitHub 저장소에 복사해서 바로 사용할 수 있는 연구 문서 템플릿입니다.

추천 저장소 구조

```
finedust-research/  
├─ README.md  
├─ .gitignore  
├─ data/  
│   ├─ raw/  
│   └─ processed/  
├─ notebooks/  
├─ src/  
├─ figures/  
├─ papers/  
│   ├─ paper_list.md  
│   ├─ reference_matrix.md  
│   └─ summaries/  
├─ notes/  
│   ├─ keyword_map.md  
│   ├─ research_questions.md  
│   └─ research_gap.md  
├─ manuscript/  
│   ├─ introduction.md  
│   ├─ related_work.md  
│   └─ outline.md  
├─ prompts/  
└─ requirements.txt
```

Paper Summary Card

Paper Summary Card

1. Bibliographic Info

- Title:
- Authors:
- Year:
- Journal/Conference:
- DOI/URL:

2. Research Question

이 논문이 해결하려는 문제는 무엇인가?

3. Data

- 대상 지역:
- 기간:
- 입력 변수:
- 출력 변수:
- 데이터 출처:

4. Method

- 사용 모델:
- 비교 모델:
- 전처리:
- 학습/검증 방법:

5. Results

- 성능 지표:
- 주요 결과:
- 가장 중요한 표/그림:

6. Contribution

이 논문의 핵심 기여는 무엇인가?

7. Limitation

이 논문의 한계는 무엇인가?

8. How I will use this paper

내 논문에서 이 논문을 어디에 사용할 것인가?

- Introduction:
- Related Work:
- Method:

- Experiment:
- Discussion:

Paper List

Paper List

No.	Title	Authors	Year	Venue	DOI/URL	Search Keyword	Screening Grade
1						A/B/C/Exclude	To read
2						A/B/C/Exclude	To read

Screening Grade

- A: 내 연구질문과 직접 연결되는 핵심 논문
- B: 배경, 방법, 데이터 중 일부가 유용한 논문
- C: 용어와 맥락 확인용 참고 논문
- Exclude: 제목은 비슷하지만 내 연구에는 쓰기 어려운 논문

Keyword Map

Keyword Map

1. My Topic

- Korean topic:
- Draft research question:
- Target pollutant:
- Target region:
- Target period:
- Method or comparison:

2. Core Keywords

Korean	English Academic Keyword	Synonyms	Notes
미세먼지	particulate matter	PM2.5, PM10, air pollution	

```
| 예측 | prediction | forecasting, estimation | |
```

3. Search Strings

```
```text
("PM2.5" OR "particulate matter") AND ("prediction" OR "forecasting") AND "machine lea
```
```

4. Search Log

```
Date	Database	Search String	Useful Papers Found	Notes
	Google Scholar			
```

Screening Log

Abstract Screening Log

```
| No. | Paper | Research Question Match | Pollutant/Data Match | Method Match | Quanti
|---:|---|---|---|---|---|---|---|
| 1 | | Yes/Partial/No | Yes/Partial/No | Yes/Partial/No | Yes/No | Introduction/Rela
```

Decision Rules

- A: 4개 이상이 Yes이고 내 연구질문과 직접 연결된다.
- B: 일부 기준만 맞지만 배경이나 방법 설명에 쓸 수 있다.
- C: 용어, 데이터 출처, 넓은 맥락 확인에만 쓴다.
- Exclude: 핵심 변수, 데이터, 방법, 연구문제가 내 주제와 다르다.

Reference Matrix

Reference Matrix

```
| No. | Paper | Problem | Data | Method | Metric | Strength | Limitation | Use in My P
|---:|---|---|---|---|---|---|---|
```

```
| 1 | | PM2.5 prediction | Weather + PM2.5 | LSTM | RMSE, MAE | | | Related Work |
| 2 | | AOD-based estimation | Satellite AOD | Random Forest | R2, RMSE | | | Metho
```

Claim Evidence Matrix

```
# Claim Evidence Matrix
```

```
Claim in My Paper	Evidence Paper	Evidence Type	Exact Support	Citation Needed		
PM2.5 예측 연구는 다양한 머신러닝 모델을 사용해 왔다.		Review / Experiment / Dataset / Poli				
고농도 이벤트 구간의 예측 오차는 별도 평가가 필요하다.				Yes	Low/Medium/High	
```

```
## How to Use
```

- Introduction의 모든 주장에 최소 1개 이상의 근거 논문을 연결한다.
- 근거가 약한 문장은 표현을 낮추거나 삭제한다.
- "아무도 하지 않았다"처럼 강한 주장은 매우 조심해서 쓴다.

Research Gap Template

```
# Research Gap
```

```
## 1. 내가 다루려는 문제
```

```
## 2. 기존 연구들이 공통적으로 한 것
```

```
## 3. 기존 연구들이 충분히 다루지 못한 것
```

```
## 4. 데이터 측면의 한계
```

```
## 5. 모델 측면의 한계
```

```
## 6. 평가 측면의 한계
```

```
## 7. 본 연구의 주장
```

기존 연구들은 [기존 연구의 초점]에 집중했지만, [부족한 부분]은 충분히 분석하지 않았다. 따라서 본 연구는 [대상 대

Introduction 5문단 템플릿

Introduction Draft

Paragraph 1. Background

미세먼지/PM2.5 문제가 사회적으로 중요한 이유를 쓴다.

Paragraph 2. Need

미세먼지 예측, 추정, 분석이 왜 필요한지 쓴다.

Paragraph 3. Previous Studies

기존 연구들이 어떤 데이터와 방법을 사용했는지 요약한다.

Paragraph 4. Gap

기존 연구의 한계를 정리한다.

Paragraph 5. Our Study

본 연구의 목적, 방법, 기여점을 쓴다.

Reproducibility Checklist

Reproducibility Checklist

- [] 데이터 출처가 명시되어 있는가?
- [] 원본 데이터와 전처리 데이터가 구분되어 있는가?
- [] 전처리 코드가 있는가?
- [] 분석 코드가 순서대로 실행되는가?
- [] `requirements.txt`가 있는가?
- [] Figure 생성 코드가 있는가?
- [] 논문에 사용된 수치가 코드 결과와 일치하는가?
- [] README만 읽고 실행할 수 있는가?

the 1990s, the number of people in the UK who are employed in the public sector has increased by 1.5 million, from 2.5 million in 1980 to 4 million in 1999. The public sector has become a major employer in the UK, and its growth has been a key factor in the overall growth of the economy.

The public sector has also become a major provider of social services, and its growth has been a key factor in the overall growth of the economy. The public sector has become a major provider of social services, and its growth has been a key factor in the overall growth of the economy.

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